AERIS-F411

Complete component and pin-connection netlist (99 parts) — ERC-reviewed

Rev A3. Five errors found and fixed on manual ERC review of the previous revision: (1) USB ESD (U2) was wired after the series resistors instead of directly at the connector — swapped so protection sits at J1 before R4/R5. (2) INA219 had no actual current-sense element — added R26, a dedicated 10mOhm shunt resistor, with VIN+/VIN- now measuring across it directly instead of across the fuse+TVS combination. (3) MCP2515 TX0RTS/TX1RTS/TX2RTS are active-low inputs and were tied to GND (constantly requesting transmission) — corrected to +3V3 (inactive). (4) R16-R19 (ESC signal lines) were wired as pull-ups to 3V3, contradicting the intended pulldown-to-GND safety behavior discussed earlier — corrected to pull down to GND. (5) U1 TPS563201 was missing its BOOT pin connection to C4 in the pin table — added. Also added a note for SN65HVD230 VREF (unused, NC). Part count is now 99 with the addition of R26. This also still includes the earlier correction: STM32F411RET6 has no VDDA pin on LQFP-64 (fixed via C13/C33 instead of a ferrite bead), and no bxCAN peripheral (fixed via the MCP2515 SPI-CAN controller).

U3 STM32F411RET6 — complete pin map (all 64 pins)

This is the single place to look up any MCU pin. All connections here match the per-component tables in the sections that follow.

Pin #	Pin name	Connects to
1	VBAT	+3V3 bus
2	PC13	Spare — do not use to drive an LED (3mA/2MHz limit)
3	PC14-OSC32_IN	X2 pin1
4	PC15-OSC32_OUT	X2 pin2
5	PH0-OSC_IN	X1 pin1
6	PH1-OSC_OUT	X1 pin2
7	NRST	SW1 pin1, C15 pin1, J3 pin5
8	PC0	Spare, J8 expansion
9	PC1	Spare, J8 expansion
10	PC2	Spare, J8 expansion
11	PC3	Spare, J8 expansion
12	VSSA	GND, C13 pin2, C33 pin2
13	VREF+	+3V3 bus, C13 pin1, C33 pin1
14	PA0-WKUP	SW3 pin1, R20 pin2
15	PA1	Spare / U4 INT1 (optional)
16	PA2 / USART2_TX	J7 GPS RX pin
17	PA3 / USART2_RX	J7 GPS TX pin
18	VSS	GND
19	VDD	+3V3 bus, C8
20	PA4 / SPI1_NSS	U4 CS, R11 pin2
21	PA5 / SPI1_SCK	SPI1 bus: U4, U5, J4, U13 SCK/CLK
22	PA6 / SPI1_MISO	SPI1 bus: U4, U5, J4, U13 MISO/DO/SO
23	PA7 / SPI1_MOSI	SPI1 bus: U4, U5, J4, U13 MOSI/DI/SI
24	PC4	U5 CS, R9 pin2
25	PC5	J4 CS, R10 pin2
26	PB0	U13 CS
27	PB1	U13 INT
28	PB2	U11 DE and RE# (tied together)
29	PB10	Spare, J8/J9 expansion
30	VCAP_1	C14 pin1 (required)
31	VSS	GND
32	VDD	+3V3 bus, C9
33	PB12	LED3 R anode via R23
34	PB13	LED3 G anode via R24
35	PB14	LED3 B anode via R25
36	PB15	LED2 anode via R22
37	PC6 / USART6_TX	U11 DI

Pin #	Pin name	Connects to
38	PC7 / USART6_RX	U11 RO
39	PC8 / TIM3_CH3	J12 ESC3 signal via R18
40	PC9 / TIM3_CH4	J13 ESC4 signal via R19
41	PA8	Spare, J9 expansion
42	PA9 / USART1_TX	Debug console TX
43	PA10 / USART1_RX	Debug console RX
44	PA11 / USB_FS_DM	R4 pin2 / U2 IO1
45	PA12 / USB_FS_DP	R5 pin2 / U2 IO2
46	PA13 / SWDIO	J3 pin2
47	VSS	GND
48	VDD	+3V3 bus, C10
49	PA14 / SWCLK	J3 pin4
50	PA15	Spare, J8/J9 expansion
51	PC10	Spare, J8/J9 expansion
52	PC11	Spare, J8/J9 expansion
53	PC12	Spare, J8/J9 expansion
54	PD2	Spare, J8/J9 expansion (only PD pin on LQFP64)
55	PB3 / SWO	J3 pin6 (optional trace pin)
56	PB4	Spare, J8/J9 expansion
57	PB5	Spare, J8/J9 expansion
58	PB6 / TIM4_CH1	J10 ESC1 signal via R16
59	PB7 / TIM4_CH2	J11 ESC2 signal via R17
60	BOOT0	SW2 pin1, R8 pin1
61	PB8 / I2C1_SCL	I2C1 bus: U6, U7, U10, R13
62	PB9 / I2C1_SDA	I2C1 bus: U6, U7, U10, R12
63	VSS	GND
64	VDD	+3V3 bus, C11

1. Power section (26 parts)

Ref	Part / value	Pin	Connects to
J1	USB-C receptacle	VBUS	D2 anode
		GND / SHIELD	GND net
		CC1	R6 pin1
		CC2	R7 pin1
		D+	U2 IO1 (ESD directly at connector)
		D-	U2 IO2 (ESD directly at connector)
		SBU1, SBU2	NC
J2	XT30 connector	BATT+	Q1 Source
		BATT-	GND net
Q1	SI2333 P-FET	Source	J2 BATT+
		Drain	F1 pin1
		Gate	R1 pin1
R1	10k	pin1	Q1 Gate
		pin2	GND
F1	Fuse 2A PPTC	pin1	Q1 Drain
		pin2	D1 cathode / R26 pin1 (shunt)
D1	SMBJ12A TVS	cathode	F1 pin2 / R26 pin1 node
		anode	GND
D2	B0530W schottky	anode	J1 VBUS
		cathode	+5V bus
R26	10mOhm 1% shunt	pin1	F1 pin2 / D1 cathode (battery side)
		pin2	C1,C2 pin1 / U1 VIN node (load side)
C1	10uF	pin1	R26 pin2 (buck VIN)
		pin2	GND
C2	10uF	pin1	R26 pin2 (buck VIN)
		pin2	GND
U1	TPS563201 buck	VIN	R26 pin2 / C1,C2
		BOOT	C4 pin1
		SW	L1 pin1
		GND	GND
		FB	R2/R3 junction
		EN	VIN node (tied high)
L1	4.7uH inductor	pin1	U1 SW
		pin2	+5V bus (C3, D2 cathode, U9 IN)
C3	22uF	pin1	+5V bus
		pin2	GND
R2	20k	pin1	+5V bus
		pin2	R3 pin1 (FB node, to U1 FB)
R3	6.8k	pin1	R2 pin2 (FB node)

Ref	Part / value	Pin	Connects to
		pin2	GND
C4	100nF	pin1	U1 BOOT pin
		pin2	U1 SW node
U2	USBLC6-2SC6 ESD	IO1	J1 D+ (protection right at the connector)
		IO2	J1 D- (protection right at the connector)
		GND	GND
R4	22R	pin1	U2 IO1
		pin2	MCU U3 pin44 (PA11 / USB_DM)
R5	22R	pin1	U2 IO2
		pin2	MCU U3 pin45 (PA12 / USB_DP)
R6	5.1k	pin1	J1 CC1
		pin2	GND
R7	5.1k	pin1	J1 CC2
		pin2	GND
C5	10uF	pin1	+5V bus
		pin2	GND
U9	AMS1117-3.3 LDO	IN	+5V bus
		GND	GND
		OUT	+3V3 bus
C6	22uF	pin1	+3V3 bus
		pin2	GND
U10	INA219	VIN+	R26 pin1 (shunt, battery side)
		VIN-	R26 pin2 (shunt, load side)
		VCC	+3V3 bus
		GND	GND
		SDA	I2C1_SDA net (PB9)
		SCL	I2C1_SCL net (PB8)
		A0, A1	GND (default address)
C7	100nF	pin1	U10 VCC
		pin2	GND

2. MCU core (U3 support circuit, 20 parts)

Ref	Part / value	Pin	Connects to
C8	100nF	pin1	U3 pin19 (VDD)
		pin2	GND
C9	100nF	pin1	U3 pin32 (VDD)
		pin2	GND
C10	100nF	pin1	U3 pin48 (VDD)
		pin2	GND
C11	100nF	pin1	U3 pin64 (VDD)
		pin2	GND
C12	4.7uF	pin1	+3V3 bus (bulk, near VDD ring)
		pin2	GND
C13	1uF	pin1	U3 pin13 (VREF+)
		pin2	U3 pin12 (VSSA)
C33	100nF	pin1	U3 pin13 (VREF+)
		pin2	U3 pin12 (VSSA)
C14	2.2uF	pin1	U3 pin30 (VCAP_1) — required
		pin2	GND
SW1	Reset button	pin1	U3 pin7 (NRST)
		pin2	GND
C15	100nF	pin1	U3 pin7 (NRST)
		pin2	GND
SW2	BOOT0 button	pin1	U3 pin60 (BOOT0)
		pin2	+3V3 bus
R8	10k	pin1	U3 pin60 (BOOT0)
		pin2	GND
X1	8MHz crystal	pin1	U3 pin5 (PH0-OSC_IN)
		pin2	U3 pin6 (PH1-OSC_OUT)
C16	10pF	pin1	X1 pin1 node
		pin2	GND
C17	10pF	pin1	X1 pin2 node
		pin2	GND
X2	32.768kHz crystal	pin1	U3 pin3 (PC14-OSC32_IN)
		pin2	U3 pin4 (PC15-OSC32_OUT)
C18	9pF	pin1	X2 pin1 node
		pin2	GND
C19	9pF	pin1	X2 pin2 node
		pin2	GND
J3	SWD header 6-pin	pin1 (VDD)	+3V3 bus
		pin2 (SWDIO)	U3 pin46 (PA13)
		pin3 (GND)	GND

Ref	Part / value	Pin	Connects to
		pin4 (SWCLK)	U3 pin49 (PA14)
		pin5 (NRST)	U3 pin7 (NRST)
		pin6 (SWO)	U3 pin55 (PB3), optional

3. Memory / storage (7 parts)

Ref	Part / value	Pin	Connects to
U5	W25Q64JV flash	CS	R9 pin2 (also U3 pin24, PC4)
		CLK	SPI1_SCK net (U3 pin21, PA5)
		DI	SPI1_MOSI net (U3 pin23, PA7)
		DO	SPI1_MISO net (U3 pin22, PA6)
		WP#	+3V3 bus
		HOLD#/RESET#	+3V3 bus
		VCC / GND	+3V3 bus / GND
C20	100nF	pin1	U5 VCC
		pin2	GND
R9	10k	pin1	+3V3 bus
		pin2	U5 CS / U3 pin24 (PC4)
J4	microSD socket	CS/DAT3	R10 pin2 (also U3 pin25, PC5)
		CMD/MOSI	SPI1_MOSI net (PA7)
		CLK	SPI1_SCK net (PA5)
		DAT0/MISO	SPI1_MISO net (PA6)
		DAT1, DAT2	NC
		VDD / VSS	+3V3 bus / GND
C21	100nF	pin1	J4 VDD
		pin2	GND
C22	1uF	pin1	J4 VDD
		pin2	GND
R10	10k	pin1	+3V3 bus
		pin2	J4 CS / U3 pin25 (PC5)

4. Sensors (11 parts)

Ref	Part / value	Pin	Connects to
U4	ICM-42688-P IMU	VDD, VDDIO	+3V3 bus
		GND	GND
		CS	R11 pin2 / U3 pin20 (PA4, SPI1_NSS)
		SCLK	SPI1_SCK net (PA5)
		SDI	SPI1_MOSI net (PA7)
		SDO	SPI1_MISO net (PA6)
		INT1	U3 pin15 (PA1), optional
		FSYNC	GND (unused)
C23	100nF	pin1	U4 VDD
		pin2	GND
R11	10k	pin1	+3V3 bus
		pin2	U4 CS / U3 pin20 (PA4)

Ref	Part / value	Pin	Connects to
U6	BME280	VDD / GND	+3V3 bus / GND
		SCK	I2C1_SCL net (U3 pin61, PB8)
		SDI	I2C1_SDA net (U3 pin62, PB9)
		CSB	+3V3 bus (I2C mode)
		SDO	GND (addr 0x76) or +3V3 (0x77)
C24	100nF	pin1	U6 VDD
		pin2	GND
U7	DS3231M RTC	VCC / GND	+3V3 bus / GND
		SDA	I2C1_SDA net (PB9)
		SCL	I2C1_SCL net (PB8)
		VBAT	BT1 pin+
		INT/SQW	Spare GPIO or NC
C25	100nF	pin1	U7 VCC
		pin2	GND
C26	100nF	pin1	U7 VBAT
		pin2	GND
BT1	CR1220 holder	pin+	U7 VBAT
		pin-	GND
R12	4.7k	pin1	+3V3 bus
		pin2	I2C1_SDA net (PB9)
R13	4.7k	pin1	+3V3 bus
		pin2	I2C1_SCL net (PB8)

5. Communication interfaces (16 parts)

Ref	Part / value	Pin	Connects to
U13	MCP2515 CAN ctrl	VDD / VSS	+3V3 bus / GND
		CS	U3 pin26 (PB0)
		SCK	SPI1_SCK net (PA5)
		SI	SPI1_MOSI net (PA7)
		SO	SPI1_MISO net (PA6)
		INT	U3 pin27 (PB1)
		OSC1, OSC2	X3 pins 1, 2
		TX0/1/2RTS	+3V3 bus (active-low inputs, tie high = inactive)
		RX0BF, RX1BF	NC
		TXCAN	U8 pin D
		RXCAN	U8 pin R
C27	100nF	pin1	U13 VDD
		pin2	GND
X3	8MHz crystal	pin1	U13 OSC1
		pin2	U13 OSC2
C28	22pF	pin1	X3 pin1 node
		pin2	GND
C29	22pF	pin1	X3 pin2 node
		pin2	GND
U8	SN65HVD230	VCC / GND	+3V3 bus / GND
		D	U13 TXCAN
		R	U13 RXCAN
		Rs	GND (high-speed mode)
		CANH, CANL	J5 pins
		VREF	NC (internal bias reference output, unused)
C30	100nF	pin1	U8 VCC
		pin2	GND
R14	120R (DNP)	pin1	J5 CANH
		pin2	J5 CANL
J5	CANH/CANL header	CANH	U8 CANH / R14 pin1
		CANL	U8 CANL / R14 pin2
U11	MAX3485	VCC / GND	+3V3 bus / GND
		DI	U3 pin37 (PC6, USART6_TX)
		RO	U3 pin38 (PC7, USART6_RX)
		DE, RE#	U3 pin28 (PB2), both tied together
		A, B	J6 pins
C31	100nF	pin1	U11 VCC
		pin2	GND
R15	120R (DNP)	pin1	J6 A

Ref	Part / value	Pin	Connects to
		pin2	J6 B
J6	A/B header	A	U11 A / R15 pin1
		B	U11 B / R15 pin2
J7	GPS header	TX (module RX)	U3 pin16 (PA2, USART2_TX)
		RX (module TX)	U3 pin17 (PA3, USART2_RX)
		VCC / GND	+3V3 bus / GND
J8	GPIO expansion	multiple	Spare GPIOs: PC0-3, PB10, PA8, PA15, PC10-12, PD2
J9	PWM expansion	multiple	Spare timer-capable GPIOs as needed

6. Connectors and indicators (19 parts)

Ref	Part / value	Pin	Connects to
J10	ESC1 pad	SIGNAL	U3 pin58 (PB6, TIM4_CH1) / R16 pin1
		GND	GND
R16	10k	pin1	J10 SIGNAL
		pin2	GND (pulldown, prevents false arming on reset)
J11	ESC2 pad	SIGNAL	U3 pin59 (PB7, TIM4_CH2) / R17 pin1
		GND	GND
R17	10k	pin1	J11 SIGNAL
		pin2	GND (pulldown)
J12	ESC3 pad	SIGNAL	U3 pin39 (PC8, TIM3_CH3) / R18 pin1
		GND	GND
R18	10k	pin1	J12 SIGNAL
		pin2	GND (pulldown)
J13	ESC4 pad	SIGNAL	U3 pin40 (PC9, TIM3_CH4) / R19 pin1
		GND	GND
R19	10k	pin1	J13 SIGNAL
		pin2	GND (pulldown)
SW3	User button	pin1	U3 pin14 (PA0) / R20 pin2
		pin2	GND
R20	10k	pin1	+3V3 bus
		pin2	SW3 pin1
C32	100nF	pin1	SW3 pin1
		pin2	GND
LED1	Power LED, green	anode	+3V3 bus via R21
		cathode	GND
R21	330R	pin1	+3V3 bus
		pin2	LED1 anode
LED2	Boot/activity LED	anode	R22 pin2
		cathode	GND
R22	330R	pin1	U3 pin36 (PB15)
		pin2	LED2 anode
LED3	RGB LED, common cathode	R anode	R23 pin2
		G anode	R24 pin2
		B anode	R25 pin2
		common cathode	GND
R23	220R	pin1	U3 pin33 (PB12)
		pin2	LED3 R anode
R24	330R	pin1	U3 pin34 (PB13)
		pin2	LED3 G anode
R25	330R	pin1	U3 pin35 (PB14)

Ref	Part / value	Pin	Connects to
		pin2	LED3 B anode

7. Full sourcing table — every designator

KiCad = ships in official libraries, no download needed. SnapEDA = export from snapeda.com in KiCad format (bundles symbol + footprint + 3D model together).

Ref	Value / exact part	Source	Footprint
J1	USB-C receptacle 16P	KiCad	USB_C_Receptacle_GCT_USB4085
J2	XT30PW-M	SnapEDA: "XT30PW-M"	from SnapEDA export
Q1	SI2333	SnapEDA: "SI2333"	SOT-23
R1	10k	KiCad	R_0603_1608Metric
F1	Fuse 2A PPTC	KiCad	Fuse_1206_3216Metric
D1	SMBJ12A	KiCad	D_SMB
D2	B0530W	SnapEDA: "B0530W"	SOD-123
C1	10uF	KiCad	C_0805_2012Metric
C2	10uF	KiCad	C_0805_2012Metric
U1	TPS563201	SnapEDA: "TPS563201"	from SnapEDA export
L1	4.7uH	KiCad	L_1210_3225Metric
C3	22uF	KiCad	C_0805_2012Metric
R2	20k	KiCad	R_0603_1608Metric
R3	6.8k	KiCad	R_0603_1608Metric
C4	100nF	KiCad	C_0603_1608Metric
U2	USBLC6-2SC6	KiCad	SOT-23-6
R4	22R	KiCad	R_0603_1608Metric
R5	22R	KiCad	R_0603_1608Metric
R6	5.1k	KiCad	R_0603_1608Metric
R7	5.1k	KiCad	R_0603_1608Metric
C5	10uF	KiCad	C_0805_2012Metric
U9	AMS1117-3.3	KiCad	SOT-223-3_TabPin2
C6	22uF	KiCad	C_0805_2012Metric
U10	INA219	SnapEDA: "INA219"	from SnapEDA export
R26	10mOhm 1% shunt	SnapEDA: "current sense resistor 10mohm" or KiCad Device:R_Shunt	R_2512_6332Metric (higher power rating)
C7	100nF	KiCad	C_0603_1608Metric
U3	STM32F411RET6	KiCad	LQFP-64_10x10mm_P0.5mm
C8	100nF	KiCad	C_0603_1608Metric
C9	100nF	KiCad	C_0603_1608Metric
C10	100nF	KiCad	C_0603_1608Metric
C11	100nF	KiCad	C_0603_1608Metric
C12	4.7uF	KiCad	C_0805_2012Metric
C13	1uF	KiCad	C_0603_1608Metric
C33	100nF	KiCad	C_0603_1608Metric
C14	2.2uF X5R	KiCad	C_0805_2012Metric
SW1	SMD tactile switch	KiCad	SW_SPST_B3U-1000P
SW2	SMD tactile switch	KiCad	SW_SPST_B3U-1000P

Ref	Value / exact part	Source	Footprint
SW3	SMD tactile switch	KiCad	SW_SPST_B3U-1000P
C15	100nF	KiCad	C_0603_1608Metric
R8	10k	KiCad	R_0603_1608Metric
X1	8MHz crystal	KiCad	Crystal_SMD_3225-4Pin_3.2x2.5mm
X3	8MHz crystal	KiCad	Crystal_SMD_3225-4Pin_3.2x2.5mm
C16	10pF	KiCad	C_0603_1608Metric
C17	10pF	KiCad	C_0603_1608Metric
X2	32.768kHz crystal	KiCad	Crystal_SMD_3215-2Pin_3.2x1.5mm
C18	9pF	KiCad	C_0603_1608Metric
C19	9pF	KiCad	C_0603_1608Metric
J3	SWD header 6-pin	SnapEDA: "TC2030-IDC-NL" (or plain header)	from SnapEDA / generic
U5	W25Q64JVSSIQ	KiCad	SOIC-8_3.9x4.9mm_P1.27mm
C20	100nF	KiCad	C_0603_1608Metric
R9	10k	KiCad	R_0603_1608Metric
R10	10k	KiCad	R_0603_1608Metric
J4	microSD push-push	KiCad	microSD_HC_Molex_104031
C21	100nF	KiCad	C_0603_1608Metric
C22	1uF	KiCad	C_0603_1608Metric
U4	ICM-42688-P	SnapEDA: "ICM-42688-P"	from SnapEDA export
C23	100nF	KiCad	C_0603_1608Metric
R11	10k	KiCad	R_0603_1608Metric
U6	BME280	SnapEDA: "BME280"	from SnapEDA export
C24	100nF	KiCad	C_0603_1608Metric
U7	DS3231MZ	SnapEDA: "DS3231MZ"	from SnapEDA export
C25	100nF	KiCad	C_0603_1608Metric
C26	100nF	KiCad	C_0603_1608Metric
BT1	CR1220 holder	KiCad	BatteryHolder_Keystone_1058
R12	4.7k	KiCad	R_0603_1608Metric
R13	4.7k	KiCad	R_0603_1608Metric
U13	MCP2515	SnapEDA: "MCP2515"	from SnapEDA export
C27	100nF	KiCad	C_0603_1608Metric
C28	22pF	KiCad	C_0603_1608Metric
C29	22pF	KiCad	C_0603_1608Metric
U8	SN65HVD230	SnapEDA: "SN65HVD230"	from SnapEDA export
C30	100nF	KiCad	C_0603_1608Metric
R14	120R (DNP)	KiCad	R_0603_1608Metric
J5	2-pin header	KiCad	PinHeader_2.54mm 1x02
J6	2-pin header	KiCad	PinHeader_2.54mm 1x02
U11	MAX3485	SnapEDA: "MAX3485"	from SnapEDA export
C31	100nF	KiCad	C_0603_1608Metric

Ref	Value / exact part	Source	Footprint
R15	120R (DNP)	KiCad	R_0603_1608Metric
J7	4-pin header	KiCad	PinHeader_2.54mm 1x04
J8	Generic headers	KiCad	PinHeader_2.54mm 1xN
J9	Generic headers	KiCad	PinHeader_2.54mm 1xN
J10	3-pin pads	KiCad	PinHeader_2.54mm 1x03
J11	3-pin pads	KiCad	PinHeader_2.54mm 1x03
J12	3-pin pads	KiCad	PinHeader_2.54mm 1x03
J13	3-pin pads	KiCad	PinHeader_2.54mm 1x03
R16	10k	KiCad	R_0603_1608Metric
R17	10k	KiCad	R_0603_1608Metric
R18	10k	KiCad	R_0603_1608Metric
R19	10k	KiCad	R_0603_1608Metric
R20	10k	KiCad	R_0603_1608Metric
C32	100nF	KiCad	C_0603_1608Metric
LED1	0603 LED	KiCad	LED_0603_1608Metric
LED2	0603 LED	KiCad	LED_0603_1608Metric
R21	330R	KiCad	R_0603_1608Metric
R22	330R	KiCad	R_0603_1608Metric
LED3	RGB common cathode	KiCad	LED_RGB_1210_3216Metric
R23	220R	KiCad	R_0603_1608Metric
R24	330R	KiCad	R_0603_1608Metric
R25	330R	KiCad	R_0603_1608Metric

8. KiCad-specific build notes (missing from earlier revisions)

These are mechanical KiCad requirements, not circuit corrections — without them, ERC will throw false-positive errors even though the design is electrically correct.

8.1 PWR_FLAG requirement

KiCad's ERC requires every power net to be driven by at least one pin explicitly marked as a "Power output" pin type. Connector pins (J1 VBUS, J2 BATT+) are typically marked as passive, not power output — so ERC will flag "power pin not driven" on +5V, +BATT, and +3V3 even though the nets are correctly wired. Fix: place a PWR_FLAG symbol (power.kicad_sym:PWR_FLAG) on each of these nets once per sheet where they appear as a source: +BATT (on the power sheet), +5V (on the power sheet), +3V3 (on the power sheet, propagates via global label), and GND (one PWR_FLAG is enough board-wide, usually placed on the power sheet). This matches the PWR_FLAG symbols visible in the reference schematic you uploaded earlier — same fix, same reason.

8.2 Canonical net label names

Use these exact strings as global labels so every sheet ties together correctly — a typo like "+3v3" vs "+3V3" creates two separate nets in KiCad silently.

Net	Label to use	Scope
Battery input	+BATT	Global label — power sheet only, feeds U1 VIN path
Regulated 5V	+5V	Global label — power sheet, USB, U9 input
Regulated 3.3V	+3V3	Global label — every sheet
Ground	GND	Global label (or the GND power symbol) — every sheet
SPI1 clock	SPI1_SCK	Hierarchical label — MCU core, memory, sensors, comms sheets
SPI1 data	SPI1_MOSI, SPI1_MISO	Hierarchical label — same sheets as above
I2C1 bus	I2C1_SCL, I2C1_SDA	Hierarchical label — MCU core, sensors, power (INA219) sheets
USART2 (GPS)	GPS_TX, GPS_RX	Hierarchical label — MCU core, interfaces sheets
USART1 (debug)	DEBUG_TX, DEBUG_RX	Hierarchical label — MCU core sheet, or local if header is on same sheet
USART6 (RS485)	RS485_TX, RS485_RX	Hierarchical label — MCU core, interfaces sheets
USB data	USB_DM, USB_DP	Hierarchical label — MCU core, USB/power sheets

8.3 Global labels vs hierarchical sheet pins

Power nets (+3V3, +5V, +BATT, GND) should be global labels — they legitimately need to reach every sheet. Everything else (SPI1_SCK, I2C1_SCL, GPS_TX, etc.) should travel between sheets as proper hierarchical sheet pins, not global labels. Using global labels for signal nets works electrically but defeats the point of having separate sheets — it becomes hard to see at a glance which sheets actually talk to each other. Correct approach: on the top-level block diagram sheet, each hierarchical sheet symbol gets pins named to match (e.g. the MCU core sheet symbol gets an output pin "SPI1_SCK"), then wire between sheet symbols on the top-level sheet.

8.4 No-connect flags

Every pin marked NC in the tables above needs an explicit no-connect flag (X symbol, keyboard shortcut Q in KiCad eeschema) placed on it, or ERC will flag it as an unconnected pin warning. This applies to: J1 SBU1/SBU2, J4 DAT1/DAT2, U3 pin2 (PC13, if left unused), U8 VREF, U13 RX0BF/RX1BF, and any spare expansion GPIOs on J8/J9 you don't wire to anything yet.

8.5 DNP (do not populate) marking

R14 and R15 (CAN and RS485 termination) should be placed on the schematic and footprint both, but marked DNP in the symbol properties (right-click the symbol, Properties, check "Exclude from bill of materials" and "Exclude from board" is wrong — you DO want the footprint on the board, just not populated at assembly. The correct KiCad approach: leave "Exclude from board" unchecked, but add a custom field or BOM note "DNP" so your assembly house knows not to place it, since whether it's needed depends on whether this board ends up as a bus end-node.